

Sequence effects at independence under partial adoption

Setting. The prior covariance is $c = 0$, so the two attributes are believed unrelated, and the adoption weight satisfies $0 < \omega < 1$, so the cue read second is adopted in part. At $c = 0$ the association channel, sequence dependence carried through the prior association, is switched off. What remains is the position channel, the weakening of whichever cue is read second, which is at full strength here. Full adoption ($\omega = 1$) is the setting of Sections 4 and 5, where the position channel is absent and the association channel vanishes at $c = 0$ (Proposition IMM). All quantities below are exact at $c = 0$; the $\mathcal{O}(c)$ corrections are in the script and are not the point.

Table 1: What can be said at $c = 0$ and $0 < \omega < 1$.

	Statistic	Value at $c = 0$	What it says
<i>Between the two sequences, one evaluator</i>			
1	marginal probabilities	$P_{AB}^J(A=1) - P_{BA}^J(A=1) = (1 - \omega)(\alpha - q_0)$ $P_{AB}^J(B=1) - P_{BA}^J(B=1) = -(1 - \omega)(\beta - r_0)$	The marginals carry the whole effect. Each is the distance the cue would have moved its marginal, scaled by how much of it was not adopted. Zero only at $\omega = 1$ or when the cue delivers the prior marginal. The mechanism is position, not association: in AB the letter is the weakened cue, in BA the credential is.
2	believed association (cross-product; odds ratio, Yule's Q)	$\text{assoc}(P_{AB}^J) - \text{assoc}(P_{BA}^J) = 0$ odds ratio identical, for every c	Both sequences leave a product table at $c = 0$, so the cross-product is zero in each; the odds ratio equals the prior's because every damped step is a row or column rescaling (Lemma SEP). The paper's qualitative verdict survives: the marginals see the sequence, the association does not. What changes is the order, zeroth instead of first, and with it the relevance of small c .
3	the adoption weight	$\omega = 1 - \frac{P_{AB}^J(A=1) - P_{BA}^J(A=1)}{\alpha - q_0}$	Row 1 read backwards. The channel the paper does not study is the channel Proposition ADJ reads; the identification device lives entirely at this point. Requires $q_0 \neq \alpha$.
4	decision flip and loss	score gap = $\langle v, P_{AB}^J - P_{BA}^J \rangle$, a fixed multiple of row 1, independent of c	The share of evaluators whose decision flips is the mass of a band of fixed width, not $\mathcal{O}(c)$, and the surplus-weighted loss is not second order. <i>Inferred from linearity of the score; not separately verified.</i>

Pooled over a population, against the benchmark, $0 < \lambda < 1$

	Statistic	Value at $c = 0$	What it says
5	association of the mean belief	$\text{assoc}(\overline{P}_\lambda) - \text{assoc}(P^B) = -\lambda(1-\lambda)(1-\omega)^2(\alpha - q_0)(\beta - r_0)$	The mean belief acquires an association nobody holds. Every evaluator's table is independent and so is the benchmark's, but the mixture of two product tables with different marginals is not a product table. A pooled audit of whether the traits go together returns a nonzero value in a population where no one believes they do. The sign is fixed by whether the two cues move their marginals in the same direction relative to the prior. No counterpart at $\omega = 1$, where the mixture at $c = 0$ is $q \otimes r$.
6	marginals of the mean belief	$\overline{P}_\lambda(A=1) - P^B(A=1) = (1-\lambda)(1-\omega)(q_0 - \alpha)$	Zeroth order. P^B sets both marginals to their delivered credences, so it is a full-adoption object under the other reading, and the gap from it mixes two differences, the reading and the adoption weight. No sequence-free reference exists at interior ω under either reading: a Bayes-factor update that scales the second cue's factor by ω is itself order-dependent at $c = 0$ (script, row 5), because Wagner's commutation needs the same factor in either position. The between-sequence comparison of rows 1–2 needs no reference, which is one more reason to prefer Proposition ORD's form of the test.

The two settings compared. The primacy regime is taken for the interior, since with ω on the second cue every interior value weakens the later impression. Weisberg (*Commutativity or Holism?*, read in full 2026-09-27; see `literature/weisberg2009`) supplies the vocabulary. He separates the *rule* that propagates a change on a partition from the *inputs problem* of how an experience fixes the values on that partition; Lange's point is that reversing two experiences does not reverse their input values; Wagner's theorem, read as a dilemma, says that commutativity on experiences forces each experience to carry the same Bayes factor whatever background it meets, which is anti-holism; and rigidity, the holding of conditionals fixed, prevents a later discovery from ever *undercutting* earlier support, leaving only rebutting. Marks: † a row of Table 1 or of `verify_interior_omega.py`; ‡ inferred from those rows, not separately checked; ◇ a mapping onto Weisberg's terms, not a computation.

Table 2: Full adoption against partial adoption (primacy regime).

	Full adoption, $\omega = 1$	Partial or no adoption, $\omega < 1$
Where the order dependence arises ◇	In the rule (the association channel). The inputs q, r are the same in either position; order matters because two rigid steps on correlated partitions do not commute (Diaconis–Zabell).	In the inputs (the position channel). The value the second cue delivers depends on its position; the rule adds nothing new at $c = 0$.
Source in the model †	Association only. Zero at $c = 0$ (IMM).	Position, present at $c = 0$, plus association.

	Full adoption, $\omega = 1$	Partial or no adoption, $\omega < 1$
Status of the inputs $\diamond$	Order-independent. Anti-holistic at the inputs, in Weisberg's sense: the credential delivers q whether or not the letter was read.	Order-dependent. History-dependent inputs, not holism in Weisberg's sense: the import of the second cue depends on its position, not on the credence in any defeater proposition.
Weisberg's dilemma, Bayes-factor reading $\diamond$	The commutative horn. Field's α , Wagner's theorem, the benchmark P^B .	Off the commutative horn, but not on the holistic one. A position-dependent factor is order-dependent at $c = 0$ (row 5 of the script) because commutation needs the same factor in either position; the dependence is on position, not on a defeater belief.
Weisberg's dilemma, delivered-credence reading $\diamond$	On neither horn. Inputs fixed regardless of any background, more anti-holistic than Field; non-commutative because the implied factor $r_j/P^A(B=j)$ varies with what the first cue did. This is non-commutativity on input distributions, the case Lange declines and Weisberg sets aside (p. 9). The two readings differ as much as they ever do.	Non-commutative through the inputs, as under the other reading; the readings differ only in how the surviving part of the earlier impression is blended, linearly in probability or geometrically in odds.
Rigidity $\diamond$	Both settings are rigid. Rigidity preserves independence, which is IMM at $c = 0$ for every ω ; a later cue can never undercut what an earlier one implied, only rebut it, and the only rebutting route is the prior association c .	
Size of the effect $\dagger$	Small, first order in c , spread unevenly across statistics.	Blunt. $(1 - \omega)(\alpha - q_0)$ on a marginal whatever c is.
Marginals $\dagger$	Last-read exact, first-read drifts at first order (DRF).	Both differ at zeroth order between sequences.
Believed association $\dagger$	Second order, between sequences and pooled (DEC, PRO).	Zero between sequences at $c = 0$, first order in c . Pooled, nonzero at $c = 0$, $-\lambda(1 - \lambda)(1 - \omega)^2(\alpha - q_0)(\beta - r_0)$.
Odds ratio $\dagger$	Identical across sequences, every c .	Identical across sequences, every c . The one statistic that does not move.
Decisions $\ddagger$	Share of flips first order, loss second order (SHR, LOS).	Score gap fixed in c , so the share is a fixed band and the loss is not second order.
Classification by order in c $\dagger$	Is the contribution. Which statistics are protected, and why.	Is moot. Marginal-based statistics see the effect at zeroth order, the odds ratio never; there is no fine structure to classify.
Audit implication $\dagger$	False negative. An association audit passes a sequence-dependent population.	False positive as well. A pooled association audit returns a stereotype no member holds, from mixing alone.
Identification	The reading, from which marginal ignores the prior association (ORD, DRF).	The weight, from three ratings (ADJ). At its strongest here, since the position channel is what ADJ reads.

	Full adoption, $\omega = 1$	Partial or no adoption, $\omega < 1$
Literature stance $\diamond$	Diaconis–Zabell’s premise; Hawthorne’s objection inherited; Lange’s point declined.	Hawthorne answered by measurement; Hogarth–Einhorn’s anchoring is the model; Lange’s point conceded by one parameter; commutativity given up on principle, not only through the reading.

What the comparison says about scope. The paper’s contribution lives in the regime where the sequence effect is subtle, arising in the rule rather than in the inputs, existing only because the traits are believed related, and therefore small and spread unevenly across statistics. That is the regime in which an audit can be fooled by choosing the wrong statistic. In the primacy regime the effect is blunt and any marginal shows it; the question becomes not whether the sequence registers but how much of the earlier impression survived, which is a question about ω , answered by ADJ and nothing else. The two regimes are two different questions. Under full adoption the association audit fails by silence; under partial adoption it fails by fabrication as well.

Two sentences for the scope statement (approved 2026-09-27).

The position channel is a temporal bias of the observer, present whether or not the attributes are believed related, while the association channel runs through what one cue implies about the other and exists only when they are.

Full adoption is the setting in which the only source of sequence dependence is the association channel, which vanishes at $c = 0$; partial adoption adds the position channel, the weakening of whichever cue is read second, which is present at $c = 0$ and so lies outside the expansion in c on which the results rest.

Two paragraphs for Section 6

Both drafted 2026-09-27 and approved in discussion. The first names the two channels; the second turns the scope condition into opposite predictions about one procedure. They follow 6.A in the plan. Every quantity is a row of Table 1 or a proposition already in the manuscript; the decision row of Table 1, which is inferred, is not used. Rubric scoring is a new element, an extension of the hiring-panel example, and no empirical claim is made about whether rubrics work.

The two channels. Sequence dependence has two channels in this setting, and the results of Sections 4 and 5 concern one of them. The first is the association channel. A cue on one attribute moves the belief about the other through the prior association, so the second cue meets a belief the first has already changed. This effect is first order in c and vanishes at independence, where each cue sets its own marginal and touches nothing else (Proposition IMM). The second is the position channel. If the cue read second is adopted only in part, then which cue is weakened depends on the order, and the two sequences differ even at $c = 0$, by $(1 - \omega)(\alpha - q_0)$ on the competence marginal and $-(1 - \omega)(\beta - r_0)$ on the trustworthiness marginal. The position channel is not a feature of updating on delivered credences. A Bayes-factor update that gives the second cue’s factor the weight ω is order-dependent in the same way, since commutation requires the same factor in either position (Wagner 2002). Under full adoption the position channel is absent and the association channel is the only one, which is the setting of the results. Under partial adoption the position channel is present at $c = 0$ and dominates the association channel near it, so the classification by order in c no longer measures what it was built to measure; the marginals differ at zeroth order, the believed association still does not differ between sequences at $c = 0$, and the mean belief of a population mixing the two sequences acquires a cross-product association of $-\lambda(1 - \lambda)(1 - \omega)^2(\alpha - q_0)(\beta - r_0)$ that no member holds. What the paper says about the position channel is confined to Proposition ADJ, which recovers its strength from the ratings. The position channel is a temporal bias of the observer, present whether or not the attributes

are believed related, while the association channel runs through what one cue implies about the other and exists only when they are.

Opposite predictions about one procedure. Whether a procedure removes the sequence effect depends on which channel it removes. Consider rubric scoring, in which the rating on each attribute is fixed by the document on that attribute and a set scale, so that what each document delivers no longer depends on what was read first. Under full adoption the rubric removes the position channel and leaves the association channel untouched. The recorded scores carry the delivered credences in either sequence, the belief about the attribute read second agrees with its score, and the belief about the attribute read first sits at cK from its score (Proposition DRF), so the sequence effect lives in that belief and in the decision made from it, vanishes when the attributes are believed unrelated (Proposition IMM), and is invisible in the scores and in the association between them. Under partial adoption the rubric removes the position channel from the record only. The belief about the attribute read second sits at $(1 - \omega)[P^A(B=1) - r_1]$ from its score (Proposition ADJ), which at independence equals $(1 - \omega)(r_0 - \beta)$, so the two reading sequences differ even when the attributes are believed unrelated. The two settings therefore predict opposite things about the same procedure, and the difference is observable. In the rubric setting r_1 is on the record, so Proposition ADJ needs only the belief about trustworthiness after the credential alone and after both documents to say which prediction holds.

Where the prelude is expanded in the manuscript

The introduction’s prelude (approved 2026-09-27) names the two channels and says which one the paper studies:

More specifically, arrival-sequence dependence can enter through two channels, an **association** channel, in which one cue changes what the other implies and which exists only when the attributes are believed related, and a **position** channel, in which the observer weights the later cue less whatever the attributes are (HogarthEinhorn1992,Asch1946). Unlike the position channel, under which the later cue is adopted only in part and every marginal registers the sequence even when the attributes are believed unrelated, the association channel is one under which the sequence registers unevenly across statistics, because each impression is adopted in full (DiaconisZabell1982) and the position channel is absent. The current paper explores this setting.

Each clause has a quantitative home. Most already exist.

Table 3: Where each clause of the prelude gets its numbers.

Prelude clause	Quantitative form	Where it lives	Still to add
an association channel that exists only when the attributes are believed related	$P_{AB}^J - P_{BA}^J = c(\kappa R_1 - \kappa' R_2) + \mathcal{O}(c^2)$; zero at $c = 0$; zero only when each delivered credence equals the prior marginal it resets	DIV and IMM, Section 4	Optionally the forcing sentence: with impressions read as delivered credences, the effect is nonzero at first order whenever either cue moves its marginal (at 3.B’s spot, or a remark after DIV)
a position channel that weights the later cue less	the three lines of ADJ; ω from three ratings	5.B, Proposition ADJ	Nothing

Prelude clause	Quantitative form	Where it lives	Still to add
every marginal registers the sequence even when the attributes are believed unrelated	$(1 - \omega)(\alpha - q_0)$ on competence, $-(1 - \omega)(\beta - r_0)$ on trustworthiness, at $c = 0$	ADJ gives the first; the second is row 1 of the interior script and appears in the two-channel paragraph	Nothing. Decided 2026-09-27: it lives in the Section 6 two-channel paragraph, not in ADJ.
registers unevenly across statistics	ORD's row table, $c\kappa'$, $-c\kappa$, $\mathcal{O}(c^2)$ on the association; PRO's criterion; the summary table	5.B and the summary table, Section 5	Nothing
adopted in full, and the position channel is absent	last-read marginal pinned at $\omega = 1$ (ADJ endpoint); the association's sequence-effect c^1 coefficient carries $(1 - \omega)$; the pooled association distortion carries $(1 - \omega)^2$; a position-scaled Bayes-factor update commutes only at $\omega = 1$	rows 3–5 of the interior script; Table 1 and the two Section 6 paragraphs above	The two Section 6 paragraphs. They are the only place this clause is expanded.
the current paper explores this setting	the scope sentence, the two channels named, the rubric prediction	Section 6, the same two paragraphs, plus the Assumption 3 scope sentence	The Assumption 3 sentence, if the one-cue-per-attribute principle is to be stated where the assumption is

Duplication to watch. Once everything is applied, $(1 - \omega)(\alpha - q_0)$ would appear in three places: the statement of ADJ, 6.A's rival-mechanisms paragraph, and the two-channel paragraph. ADJ is the proposition and carries it; the other two cite ADJ and do not repeat the expression. The pooled association distortion is stated only in the two-channel paragraph and is fine.

What must not go in. The decision row of Table 1 (share of flips and loss at interior ω) is inferred from linearity and not checked; it stays out of the manuscript. Nothing at interior ω enters as a proposition; the interior numbers appear inside Section 6 prose, as scope justification, each traceable to a row of the interior script.

Order of application. 5.B before the two Section 6 paragraphs, since both cite ADJ; B.A before either, for Hawthorne2004.